

DUSTIN LEUNG

dustinleung07@gmail.com — 503-805-8220 — www.linkedin.com/in/dustin-leung

EDUCATION

Master of Science, Mechanical Engineering

University of Michigan

Graduate Researcher, Smart and Sustainable Automation Research Lab

Expected May 2028

Ann Arbor, MI

Bachelor of Science, Mechanical Engineering – Aerospace Option

Oregon State University

Cumulative GPA: 4.0/4.0; *summa cum laude*

March 2026

Corvallis, OR

EXPERIENCE

In-Space Manufacturing Intern

NASA Marshall Space Flight Center

June 2026 – August 2026

Huntsville, AL

- Developed a closed-loop, GPU-accelerated process monitoring system on an NVIDIA Jetson Thor for Laser Directed Energy Deposition, integrating multi-camera sensing and real-time AI inference
- Built a real-time detection system with a decoupled, multi-rate Holoscan pipeline ensuring no subsystem stalls another, streaming camera inference and LLM analytics to a physical LED display
- Implemented a retrieval-augmented generation pipeline on a Raspberry Pi using a local LLM to index and query batch documentation and manuals for LDED operations

Design Engineer – Structures & Tooling

PCC Structurals – Small Structures Business Operations

March 2026 – June 2026

Clackamas, OR

- Engineered injection-molding tools, inserts, and chills in SolidWorks and NX, improving casting yield and cutting labor hours 10% via single-shot tooling, unlocking \$2–3M in engineering revenue
- Optimized gating configurations through solidification-driven design and ProCAST simulation to minimize defect formation and improve investment-casting quality

Part & Dimensional Mechanical Engineering Intern

PCC Structurals – Small Structures Business Operations

March 2025 – September 2025

Clackamas, OR

- Applied design-for-manufacturing principles to optimize gating geometry for investment-cast aerospace components in SolidWorks and NX, unlocking development opportunities valued at over \$1M
- Conducted dimensional and defect-trend analysis on aerospace components to identify failure modes, driving Root Cause Corrective Action that reduced scrap and rework
- Selected for PCC's Intern Award for Impact and Innovation among interns across all divisions of Precision Castparts Corporation

Mechanical Design Engineering Intern

The Boeing Company – Everett Factory

March 2024 – September 2024

Everett, WA

- Designed and validated 3D-printed production aids in CATIA for the 767 production line, eliminating a recurring defect and improving reliability on safety-critical assemblies
- Directed Root Cause Corrective Action for a recurring 767 production defect, implementing solutions that reduced rework and improved first-pass build quality across multiple aircraft positions

PROJECTS & LEADERSHIP

Structures Project Lead

Oregon State University – IREC Rocketry Team

September 2024 – March 2026

Corvallis, OR

- Led a student team on structural design of secondary components, including the coupler and nose cone, integrating with the primary airframe and payload systems
- Characterized the team's in-house additive manufacturing process, testing printer capabilities and building a material property dataset to guide future component designs

TECHNICAL SKILLS

CAD & Design Software:

SolidWorks; CATIA; Siemens NX; Fusion 360; Autodesk Inventor; OnShape

Programming & Scripting:

Python; C++; MATLAB; Arduino

AI/ML & Embedded:

NVIDIA Jetson; Raspberry Pi; Holoscan; Computer Vision; LLM/RAG

Simulation & Analysis:

Ansys; ProCAST; JMP; Geomagic Design X; Polyworks; COMSOL

Fabrication & Prototyping:

3D Printing (FDM & SLA); CNC; Mill; Lathe; General Shop Tools